SQUARE MODULI, SPARSE MASKS, AND SIGN OBSTRUCTIONS FOR VANISHING ERDŐS SUMS

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Abstract. For an odd integer $q > 1$, an Erdős function modulo q takes values ± 1 away from multiples of q and vanishes on those multiples. We study the difference between vanishing of its harmonic sum and the existence of odd vectors in the rational Okada space. We prove that an odd Okada vector exists if and only if q is a square. The proof eliminates prime supports in decreasing order on the maximal 2-power quotients of unit groups, preserving explicit 2-adic congruences. It replaces the finite obstruction computations in the preceding manuscript by a general argument. More generally, for every prime ℓ, a rational Okada vector congruent to 1 modulo ℓ away from zero exists exactly when $q = m^\ell$ with $m \equiv 1 \pmod{\ell}$.

For actual sign vectors, we give exact sparse masks that enlarge the known counterexamples. Every modulus $3^{2a}5^{2b}7^{2c}$ with $a, b, c \geq 1$ carries a vanishing Erdős function; this gives a complete classification on prime support contained in $\{3, 5, 7\}$. A further mask permits an arbitrary even power of 11 to be appended. In contrast, a short analytic argument proves nonvanishing for every Erdős function modulo 195^2. We also prove that every odd square satisfies the bounded integer shadow relaxation, and give an infinite family on which no invariant sign solution exists. Thus the remaining square-modulus problem is a genuine lifting problem. The nonvanishing assertion for nonsquare moduli was announced in 2025 by Bharadwaj, Murty, and Pathak; priority for the stronger odd-vector statement requires comparison with their full manuscript.

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Date: 24 October 2026 — research draft.

1. Introduction and relation to the preceding manuscript

Put $L(1,f)=\sum_{n\ge 1} f(n)/n$. For a periodic function, convergence is equivalent to a zero period sum. An *Erdős function modulo* q has $f(0)=0$ and $f(n)\in\{-1,1\}$ for $q\nmid n$. Its zero set forces q to be its least positive period. Convergence forces q to be odd. Call q a *counterexample modulus* if such a function has $L(1,f)=0$. The preceding manuscript [3] constructs counterexamples beginning with 105^2, and supplies exact certificates at 165^2, 315^2, 495^2, and 735^2. This article leaves that manuscript unchanged.

The prime-adic extension gives an exact congruence classification beyond the binary alphabet: for every prime ℓ and modulus $q>1$, a rational Okada vector taking values in $1+\ell\mathbb{Z}_{(\ell)}$ off zero exists if and only if $q=m^\ell$ with $m\equiv 1 \pmod{\ell}$. This theorem concerns a prescribed residue class, with no restriction to values ± 1.

The main distinction is between three questions: whether an odd rational Okada vector exists, whether the retained character equations admit the orbit sums of a sign function, and whether an actual sign function satisfies the full equations. The first two questions have the same simple answer: the modulus must be a square, and every odd square works. The third question has a more restrictive answer. In particular,

$$195^2=38025 \quad \text{is a square with} \quad 2\varphi(38025)+1=37441<38025,$$

but it is not a counterexample modulus. This disproves the possible sufficiency of the elementary totient condition among odd squares.

Our positive constructions use sparse periodic functions which already have vanishing harmonic sum. Inserting an earlier counterexample into their zero positions gives independent exponent increases. These are stronger than the previously known closure under multiplying two counterexample moduli: the multipliers 25, 49, 81, and 121 are not themselves counterexample moduli.

1.1. Literature and the scope of novelty.

The necessary equations used here are due to Okada [7]; their modern rational-space formulation is given by Murty and Pathak [5], and primitive-character formulations are developed by Bharadwaj [2]. The original conjecture is recorded by Livingston [4]. For the algebraic periodic coefficients considered here, a convergent value is either zero or transcendental, by the established logarithmic-form results [1, 6].

The announcement of Siddhi Pathak's seminar on 1 October 2025 states that joint work with A. Bharadwaj and M. R. Murty settles the Erdős nonvanishing question except at perfect-square periods [8]. The corresponding work is listed as submitted on Pathak's publication page [9]. We did not obtain its full text. Consequently, the nonsquare sign-function conclusion is not claimed here as a new resolution of an open problem. Our proof establishes the stronger classification of odd $\mathbb{Z}_{(2)}$-valued Okada vectors; whether that stronger statement or the same argument occurs in the submitted work remains to be checked. The square-modulus results below are stated independently of that unpublished manuscript.

2. Okada equations and the retained character system

The Okada equations below hold for any modulus $q>1$; in the binary discussion we take q odd. Write $U_d=(\mathbb{Z}/d\mathbb{Z})^\times$ and let $M(q)$ be the multiplicative monoid generated by the primes dividing q. A rational periodic function with $f(0)=0$ and mean zero belongs to the Okada space if and only if its harmonic sum vanishes. We use the following standard forms of its defining equations [7, 5, 2]:

$$\sum_{m\in M(q)} \frac{f(am)}{m}=0 \qquad (a\in U_q), \tag{2.1}$$

$$\sum_{d\mid q,\ d>1} \left(v_p(q/d)-\frac{[p\mid d]}{p-1}\right) S_d=0 \qquad (p\mid q), \tag{2.2}$$

where $f_d(x) = f((q/d)x)$ and $S_d = \sum_{x\in U_d} f_d(x)$. The primitive-character form has, for every nonprincipal primitive Dirichlet character χ of conductor $m \mid q$, the equation

$$\sum_{m|d|q} \frac{d}{\varphi(d)} \prod_{p|d,\ p\nmid m} \left(1 - \frac{\chi(p)}{p}\right) \sum_{x\in U_d} f_d(x)\overline{\chi(x)} = 0. \tag{2.3}$$

Together with (2.2) and the mean equation, these are an equivalent criterion. We make no assertion that every displayed equation is linearly independent.

Set $R = \mathbb{Z}_{(2)}$, the rationals with odd denominator. An *odd Okada vector* means a rational function in this space with $f(0) = 0$ and $f(n) \in 1 + 2R$ for every nonzero residue. Clearing its odd denominators gives an equivalent odd-integer existence question, without a height bound. Let H_d be the odd-order subgroup of U_d and put

$$Q_d = U_d/H_d = \prod_{p|d} Q_p, \qquad Q_p \simeq C_{N_p}, \quad N_p = 2^{v_2(p-1)}, \qquad h_d = |H_d|.$$

For any function define its actual orbit sums

$$\Phi_d(y) = \sum_{x\in y} f_d(x), \qquad y \in Q_d. \tag{2.4}$$

If f is odd R-valued, each $\Phi_d(y)$ is odd, since h_d is odd. Characters factoring through Q_d have squarefree conductor. Restricting (2.3) to these characters gives the same equation with the last sum replaced by $\sum_{y\in Q_d} \Phi_d(y)\overline{\chi(y)}$. The prime and mean equations also depend only on these sums. We call this retained system the 2-*shadow system*.

Conversely, if orbit sums solve the retained system, set $f_d(x) = \Phi_d(xH_d)/h_d$. Then f is invariant under odd-order units. Every omitted character coefficient is zero by orthogonality, so the retained equations imply the full criterion. Since h_d is an odd unit in R, odd orbit sums give an odd Okada vector. This explains exactly which existence question the shadow decides. For sign functions, the possible orbit sums have the additional constraints

$$\Phi_d(y) \in \{-h_d, -h_d + 2, \ldots, h_d - 2, h_d\}. \tag{2.5}$$

Such sums can be lifted to signs within each fibre, but an arbitrary sign lift need not satisfy the omitted character equations.

3. A global obstruction for nonsquare moduli

Let $R = \mathbb{Z}_{(2)} = \mathbf{Z}_{(2)}$. For every odd prime p put $N_p = 2^{v_2(p-1)}$, so that $N_p/(p-1)$ is a unit of R. We use the shadow criterion of the preceding section.

Theorem 3.1. *An odd integer $q > 1$ admits an odd rational Okada vector if and only if q is a perfect square.*

Both directions are proved below. In particular, this theorem classifies odd vectors in the rational relaxation; it does not assert that every odd square admits a vanishing sign function.

3.1. **Compression by prime support.** Write $q = \prod_{p\in P} p^{e_p}$. For every nonempty subset $S \subseteq P$ put $Q_S = \prod_{p\in S} Q_p$, where Q_p is the cyclic group of order N_p. All shadow layers whose prime support is S have this same domain. Define their sum

$$G_S = \sum_{\substack{D|q \\ \operatorname{supp}(D)=S}} \Phi_D \quad \text{on } Q_S.$$

Since the summands are odd and their number is $\prod_{p\in S} e_p$, the function G_S has constant parity

$$G_S(x) \equiv \prod_{p\in S} e_p \pmod{2R}. \tag{3.1}$$

In particular, $G_{\{p\}}(x) \equiv e_p \pmod{2R}$.

For a character χ primitive on Q_M, where $\varnothing \neq M \subseteq P$, extend χ to each Q_S with $S \supseteq M$ by projection and write $\widehat{G}_S(\chi) = \sum_{x\in Q_S} G_S(x)\overline{\chi(x)}$. Dividing the character equation by $\prod_{p\in M} p/(p-1)$ gives

$$\sum_{S\supseteq M} \prod_{p\in S\setminus M} \frac{p-\chi(p)}{p-1}\widehat{G}_S(\chi) = 0. \tag{3.2}$$

For every $p \in P$ define

$$T_p(G) = \sum_{S\ni p} \sum_{x\in Q_S} G_S(x).$$

The mean equation converts the original p-equation to

$$\sum_{\substack{D\mid q\\ p\mid D}} \big((p-1)v_p(D)+1\big) \sum_{x\in Q_D} \Phi_D(x) = 0.$$

Every layer sum belongs to $2R$, since Q_D has even cardinality and every entry of Φ_D is odd. Consequently

$$T_p(G) = -(p-1)\sum_{\substack{D\mid q\\ p\mid D}} v_p(D) \sum_{x\in Q_D} \Phi_D(x) \in 2N_pR. \tag{3.3}$$

3.2. **An integral elimination step.**

Lemma 3.2 (Decomposition with controlled parity)**.** *Let $|S| \geq 2$ and let $F : Q_S \to R$ have constant parity. If its Fourier coefficients vanish for every character nontrivial on each factor Q_p, $p \in S$, then there are functions $h_p : Q_{S\setminus\{p\}} \to R$ of constant parity such that*

$$F(x) = \sum_{p\in S} h_p(x_{S\setminus\{p\}}).$$

Proof. Fix the identity element as base point in each factor. The anchored inclusion–exclusion decomposition is

$$F(x) = \sum_{U\subseteq S} F_U(x_U), \qquad F_U(x_U) = \sum_{V\subseteq U} (-1)^{|U|-|V|} F(x_V, 0_{S\setminus V}).$$

The vanishing hypothesis says that F is a sum, over the rationals, of functions missing at least one coordinate. It therefore implies $F_S = 0$. Equivalently, this follows by applying the full mixed difference to the Fourier expansion. For $U \neq \varnothing$, all $2^{|U|}$ terms in F_U have the same parity, so F_U is even-valued. The term $F_\varnothing = F(0)$ is constant. Assign each proper subset $U \subsetneq S$ to some $p \in S \setminus U$ and inflate F_U to $Q_{S\setminus\{p\}}$. Summing the terms assigned to each p gives the required h_p. Each is even-valued except possibly the one receiving the constant term, and that one also has constant parity. □

Lemma 3.3 (Elimination circuit)**.** *Suppose $S \subseteq P$, $|S| \geq 2$, $p \in S$, and $h : Q_{S\setminus\{p\}} \to R$ has constant parity. Write $U = S \setminus \{p\}$. The changes*

$$\delta G_S = -\inf_U^S h, \qquad \delta G_U = \frac{N_p}{p-1}(p - T_p^{\mathrm{tr}})h \tag{3.4}$$

preserve every character equation (3.2)*, every congruence* (3.3)*, and the parity of* G_U*. Here* $(T_p^{\mathrm{tr}}h)(y) = h(y+\overline{p})$ *is translation by the image of* p *in* Q_U*.*

Proof. The coefficient $N_p/(p-1)$ belongs to R. Since p is odd and h has constant parity, $(p - T_p^{\mathrm{tr}})h$ is even-valued. Thus the lower-layer change is even and preserves its parity.

For a character whose primitive support contains p, the change on Q_S has zero Fourier coefficient, because it is constant in the Q_p coordinate; the lower layer does not contribute. For primitive support $M \subseteq U$, inflation multiplies its Fourier coefficient by N_p, while

$$\widehat{T_p^{\mathrm{tr}}h}(\chi) = \chi(p)\widehat{h}(\chi).$$

The two changes in (3.2), after removing the common factor indexed by $U \setminus M$, are therefore

$$-\frac{p-\chi(p)}{p-1}N_p\widehat{h}(\chi) + \frac{N_p}{p-1}(p-\chi(p))\widehat{h}(\chi) = 0.$$

All other primitive supports are unaffected.

Put $H = \sum_{x\in Q_U} h(x)$. Translation preserves sums, so the total changes in the two layers are $-N_pH$ and $+N_pH$. Consequently the total mean is unchanged, and $T_s(G)$ is unchanged for every $s \neq p$. The change in $T_p(G)$ is $-N_pH$. The set U is nonempty, so $|Q_U|$ is even; since h has constant parity, $H \in 2R$. Thus $-N_pH \in 2N_pR$, as required. □

3.3. **Proof of the global obstruction.**

Proof of Theorem 3.1, necessity. Suppose that an odd Okada vector exists. Form the compressed functions G_S above, which satisfy (3.2) and (3.3) and have the constant parities (3.1).

Process the subsets $S \subseteq P$ of size at least two in decreasing order of their size. When S is reached, all strict supersets have already been eliminated. Hence (3.2) for primitive support S says precisely that G_S has no fully primitive Fourier component. Every earlier alteration of this layer was even-valued, so it still has constant parity. Lemma 3.2 expresses it as a sum of functions missing one coordinate. Apply Lemma 3.3 to each summand. These operations replace G_S by zero, alter only smaller layers, preserve every character equation and every congruence (3.3), and preserve the parity of each smaller layer. The process terminates.

Only the singleton functions $G_{\{p\}}$ remain. Their nonprincipal Fourier coefficients vanish by (3.2), so each is a constant $a_p \in R$. Its parity is still e_p modulo $2R$. But (3.3) now gives

$$N_pa_p = T_p(G) \in 2N_pR,$$

so a_p is even, and therefore e_p is even. This holds for every $p \mid q$. Thus q is a perfect square. □

Proof of Theorem 3.1, sufficiency. Suppose every e_p is even. For each nonempty support $S \subseteq P$ choose one prime $p(S) \in S$ and a nontrivial quadratic character $\eta_{p(S)}$ on $Q_{p(S)}$. On each layer D with support S, set

$$\Phi_D(y) = (-1)^{v_{p(S)}(D)}\eta_{p(S)}(y_{p(S)}).$$

Every value is 1 or -1, and each layer sum is zero. Hence the mean and all prime equations vanish. For each fixed support S, summing over the even number $e_{p(S)}$ of possible positive exponents of $p(S)$ gives $G_S = 0$. Thus all character equations vanish as well. Shadow descent yields an odd rational Okada vector. Its original layer values are $\Phi_D/|H_D|$; their denominators are odd, but they need not have absolute value one. □

Corollary 3.4. *Every convergent Erdős sign sum at an odd nonsquare modulus is nonzero, and hence transcendental. Any vanishing Erdős sign function has an odd square modulus.*

The maximal 2-power quotient has a precise role in the argument: $N_p/(p-1)$ must be a unit of $\mathbb{Z}_{(2)}$ in the elimination circuit. Replacing Q_p by a smaller 2-quotient can introduce an even denominator and invalidate the preservation of parity.

4. Sparse zero-sum masks and inflation of counterexamples

The usual composition of two counterexamples multiplies their moduli. The construction below also permits multiplication by certain prime squares which themselves admit no counterexample. Its auxiliary functions have several zero residues, and therefore are not themselves Erdős functions.

Definition 4.1. Let $1 < m \mid Q$. A *zero-harmonic mask of type* (Q, m) is a Q-periodic function $g\colon \mathbb{Z} \to \{-1, 0, 1\}$ such that

$$g(n) = 0 \quad \Longleftrightarrow \quad m \mid n, \qquad \sum_{n=1}^{Q} g(n) = 0, \qquad L(1, g) = 0.$$

Theorem 4.2 (Mask insertion)**.** *Suppose that f is an Erdős counterexample of modulus q, and that g is a zero-harmonic mask of type (Q, m) with $Q \mid mq$. Then*

$$H(n) = \begin{cases} g(n), & m \nmid n, \\ f(n/m), & m \mid n \end{cases}$$

is an Erdős counterexample of modulus mq. In particular, a mask of type (am, m) permits multiplication by m of every counterexample modulus divisible by a.

Proof. The condition $Q \mid mq$ makes H periodic modulo mq. Its zero set is exactly $mq\mathbb{Z}$, and all other values are signs. The identity

$$H(n) = g(n) + \mathbf{1}_{m|n} f(n/m)$$

shows that its sum over a period is zero. Both series on the right converge, so the same identity gives

$$L(1, H) = L(1, g) + \frac{1}{m} L(1, f) = 0.$$

Moreover, its least positive period is mq, as is forced by its zero set. No coprimality hypothesis is needed. □

Proposition 4.3 (Three exact masks)**.** *There exist zero-harmonic masks of each of the three types in Table 1.*

Q	m	Q/m	Character orders used	Unit equations
11025	49	225	$2, 2, 2$ at $3, 5, 7$	5040
11025	25	441	$2, 4, 2$ at $3, 5, 7$	5040
99225	81	1225	$2, 2, 2$ at $3, 5, 7$	45360

Table 1. Certified masks. In each case all the indicated unit equations, the three prime equations, and the period sum vanish exactly.

Proof. The divisor-layer sign tables are the supplementary certificates `mask_11025_hole_49.json`, `mask_11025_hole_25.json`, and `mask_99225_hole_81.json`. For each divisor $d > 1$ of Q, write $n = (Q/d)x$ with $(x, d) = 1$. A table row gives $g(n)$ as a function of the discrete logarithm classes of x at the primes dividing d, listed in increasing order. The coordinates have the orders shown in Table 1. The primitive roots are $2, 2, 3$ modulo $3, 5, 7$, respectively. A missing row is identically zero.

The independent standard-library program `verify_sparse_masks.py` expands these tables to every residue, checks the advertised zero sets and signs, and verifies Okada's original equations using exact rational arithmetic. For each $p \mid Q$, it computes the resolvent

$$T_p h(a) = \sum_{j \geq 0} \frac{h(p^j a)}{p^j}$$

by solving $T_p h(a) = h(a) + T_p h(pa)/p$ on every cycle and incoming tree of the map $a \mapsto pa \pmod{Q}$. Every local recurrence is checked again. The commuting product of these resolvents gives the smooth sum for every unit residue. All $\varphi(Q)$ such values vanish. The three integer-scaled prime-valuation equations and the period sum also vanish. Consequently Okada's criterion gives $L(1, g) = 0$ in each case. This verification does not use the projected shadow equations or the integer-programming solver used to discover the tables. □

Corollary 4.4 (Three universal inflation rules)**.** *If q is a counterexample modulus, then so are*

$$\begin{aligned} &49q, \quad \text{if } 225 \mid q, \\ &25q, \quad \text{if } 441 \mid q, \\ &81q, \quad \text{if } 1225 \mid q. \end{aligned}$$

Theorem 4.5 (All square moduli on the radical 105)**.** *For every $a, b, c \geq 1$, the modulus*

$$q = 3^{2a}5^{2b}7^{2c}$$

admits an Erdős counterexample.

Proof. The previously certified seeds $105^2 = 3^2 5^2 7^2$ and $315^2 = 3^4 5^2 7^2$ give the cases $a = 1$ and $a = 2$, with $b = c = 1$. Every constructed modulus remains divisible by 225, 441, and 1225. The three rules of Corollary 4.4 therefore allow us to increase c by 1, b by 1, and a by 2, independently. The two seeds cover the two parities of a. □

Corollary 4.6 (Classification of odd $\{3, 5, 7\}$-smooth moduli)**.** *Let $q > 1$ be odd and have no prime factors outside $\{3, 5, 7\}$. Then q is a counterexample modulus if and only if it is a square divisible by 105^2.*

Proof. Theorem 4.5 proves sufficiency. The non-square obstruction proved in this paper excludes every non-square q with this prime support. If a square q omits at least one of the three primes, then

$$\frac{\varphi(q)}{q} \geq \left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = \frac{8}{15} > \frac{1}{2}.$$

Proposition 4.7, applied to an Erdős function, would require $2\varphi(q) \leq q - 1$, excluding this case. □

The inflation statements also apply to counterexamples with additional prime factors. For example, the seed 165^2 gives all moduli $165^2 \cdot 49^k$, $k \geq 0$.

4.1. **A necessary support bound.** The next estimate explains why the smaller multiplier 9 cannot be obtained from a mask of period 11025.

Proposition 4.7 (Support bound)**.** *Let h be a rational Q-periodic function with $h(0) = 0$, zero period sum, $L(1, h) = 0$, and $|h(a)| = 1$ for every unit a modulo Q. Then*

$$2\varphi(Q) \leq \sum_{n \bmod Q} |h(n)|.$$

In particular, a zero-harmonic mask of type (Q, m) can exist only if

$$2\varphi(Q) \leq Q - Q/m.$$

Proof. Write $M(Q)$ for the monoid generated by the primes dividing Q. For each unit a, the unit Okada equation and the triangle inequality give

$$1 = |h(a)| \leq \sum_{\substack{r \in M(Q) \\ r > 1}} \frac{|h(ar)|}{r}.$$

Every residue $n \pmod{Q}$ has total weight exactly 1 in the double sum

$$\sum_{a\in(\mathbb{Z}/Q\mathbb{Z})^\times} \sum_{r\in M(Q)} \frac{|h(ar)|}{r}.$$

Indeed, if $d = (n, Q)$, solutions of $ar \equiv n \pmod{Q}$ with a a unit exist exactly when $(r, Q) = d$, and their number is $\varphi(Q)/\varphi(Q/d)$. To evaluate the resulting weight prime by prime, let $p^e \| Q$ and $v = \min(v_p(n), e)$. If $v < e$, the local fibre size is p^v and the sole exponent in r is v, giving weight $p^v p^{-v} = 1$. If $v = e$, the local weight is

$$\varphi(p^e) \sum_{j\ge e} p^{-j} = (p-1)p^{e-1} \frac{p^{1-e}}{p-1} = 1.$$

All terms are nonnegative, so the rearrangement is justified. Summing the unit inequalities and removing the $r = 1$ contribution gives $\varphi(Q) \le \sum_n |h(n)| - \varphi(Q)$, as required. □

For $Q = 11025$, a mask with $m = 9$ would have support size $11025 - 1225 = 9800$, while $2\varphi(11025) = 10080$. It is therefore impossible, without any symmetry restriction. More generally, on radical 105, the support inequality forces

$$m \ge \frac{1}{1 - 2\varphi(Q)/Q} = \frac{35}{3}.$$

Among positive even powers of 3, the first permitted multiplier is 81. Thus the use of an 81-mask and two initial 3-exponents reflects a genuine support obstruction.

4.2. **An explicit 49-mask.** For completeness, the shortest certificate is displayed in Table 2. For $n = (11025/d)x$ with $(x, d) = 1$, each row lists the signs in lexicographic order of the quadratic-character coordinates at the primes dividing d. Coordinate 0 means Legendre symbol $+1$ and coordinate 1 means -1. The last coordinate varies fastest. Every omitted divisor layer is zero. The listed layers are exactly those with $7 \mid d$, so the zero set is exactly the multiples of 49.

d	Signs	d	Signs	d	Signs
7	+−	147	−+−+	735	+−−++−−+
21	−+−+	175	+−−+	1225	+−−+
35	−++−	245	+−−+	1575	+−−++−−+
49	+−	315	++−+−−−+	2205	+−−++−−+
63	−−++	441	−−++	3675	+−−++−−+
105	++−−−−++	525	+−+++−−−	11025	−++−−++−

TABLE 2. A zero-harmonic mask of type $(11025, 49)$.

4.3. **Further masks on the radical 165.** Two additional compact certificates give zero-harmonic masks of types

$$(Q, m) = (27225, 121), \qquad (Q, m) = (245025, 81).$$

Their primitive-root coordinates have orders $2, 4, 2$ at $3, 5, 11$, respectively, with primitive root 2 at all three primes. The same independent full-residue verifier checks 13200 and 118800 unit equations, respectively, as well as all prime equations, the exact zero sets, and the zero period sums. The certificates are `mask_27225_hole_121.json` and `mask_245025_hole_81.json`.

Corollary 4.8. *Every counterexample modulus q has the following further inflation rules:*

$$225 \mid q \implies 121q \text{ is a counterexample modulus},$$
$$3025 \mid q \implies 81q \text{ is a counterexample modulus}.$$

In particular, every modulus $3^{2a}5^2 11^{2c}$, $a, c \geq 1$, *admits a counterexample. More generally, the same holds for*

$$3^{2a}5^{2b}11^{2c} \qquad (a, c \geq b \geq 1).$$

Proof. Apply the mask insertion theorem. The previously certified seeds $165^2 = 3^2 5^2 11^2$ and $495^2 = 3^4 5^2 11^2$, together with the multipliers 81 and 121, give all exponents with $b = 1$. A product of b such counterexample moduli gives any prescribed $a, c \geq b$, by writing each of a and c as a sum of b positive integers and using the existing composition rule for ordinary counterexamples. □

Combining the 121 inflation rule with Theorem 4.5 also gives every modulus

$$3^{2a}5^{2b}7^{2c}11^{2d}, \qquad a, b, c \geq 1, \quad d \geq 0.$$

Thus the fourth prime can have any nonnegative even exponent, independently of the three positive even exponents in the complete radical-105 family. No claim is made here about the still-uncovered exponents with prime support $\{3, 5, 11\}$, nor about an arbitrary additional prime.

5. The modulus 195^2

The distinction between an odd Okada vector and a sign vector is already visible at $q = 195^2$. We prove the conjectured nonvanishing at this modulus for arbitrary Erdős functions, without an invariance hypothesis. The proof combines a bound on the reciprocal mass in Okada's unit equations with three character equations of conductor 13 and the prime-3 equation.

Lemma 5.1 (Weighted forcing)**.** *Let $q > 1$ be odd, let $M(q)$ be the multiplicative monoid generated by its prime divisors, and put*

$$R_q = \prod_{p \mid q} \frac{p}{p-1}, \qquad E_q = \frac{R_q(1 - 1/q) - 2}{2}.$$

Suppose that an Erdős function f modulo q has $L(1, f) = 0$. For every $a \in U_q$ the reciprocal mass of the agreeing terms is

$$\sum_{\substack{m \in M(q),\ m>1,\ q \nmid m \\ f(am)=f(a)}} \frac{1}{m} = E_q.$$

Consequently, if $m \in M(q)$, $m > 1$, $q \nmid m$ and $1/m > E_q$, then $f(am) = -f(a)$ for every $a \in U_q$.

Proof. Okada's unit equation is $\sum_{m \in M(q)} f(am)/m = 0$. The full reciprocal mass is R_q, and the terms divisible by q have mass R_q/q and contribute zero. After removing $m = 1$, the remaining nonzero terms have total mass $R_q(1 - 1/q) - 1$. Multiplying the equation by $f(a)$ shows that the mass of disagreeing terms exceeds that of agreeing terms by 1. Solving these two equations gives the asserted value E_q. An individual agreeing term cannot have mass greater than their total mass. □

Theorem 5.2. *Every convergent Erdős sum of modulus $q = 195^2 = 38025$ is nonzero.*

Proof. Assume to the contrary that $L(1, f) = 0$, and write $q = 3^2 5^2 13^2$. Here $R_q = 65/32$ and

$$E_q = \frac{1}{64} - \frac{1}{37440} < \frac{1}{64}.$$

Lemma 5.1 therefore gives

$$f(ma) = -f(a) \qquad (a \in U_q), \qquad m \in \{3, 5, 9, 13, 15, 25, 27, 39, 45\}. \tag{5.1}$$

The unit restriction. The equation for $m = 9$ implies that the restriction of f to U_q depends only on the image in $U_{q/9}$: if $a \equiv b \pmod{q/9}$, then $f(a) = -f(9a) = -f(9b) = f(b)$. Under $U_q \cong U_9 \times U_{25} \times U_{169}$ this removes the U_9 coordinate. The equations for $m = 25$ and $m = 13$

similarly remove the U_{25} coordinate and the kernel of $U_{169} \to U_{13}$. Hence $f(a) = G(a \bmod 13)$ for $a \in U_q$.

Given $a \in U_q$, choose $b \in U_q$ with $b \equiv 3a \pmod{q/9}$; this is possible by the Chinese remainder theorem, since $q/9 = 4225$ is coprime to 3. Then $9b \equiv 27a \pmod{q}$, and (5.1) gives $f(b) = f(a)$. It follows that $G(3t) = G(t)$. The element 3 has order 3 modulo 13, and 2 is a primitive root with $\langle 3 \rangle = \{2^0, 2^4, 2^8\}$. There are thus signs $F_j \in \{\pm 1\}$, indexed by $j \in \mathbb{Z}/4\mathbb{Z}$, such that

$$f(a) = F_{\mathrm{ind}_2(a \bmod 13) \bmod 4} \qquad (a \in U_q). \tag{5.2}$$

Put $D = \{1, 3, 5, 9, 13, 15, 25, 39, 45\}$. For $g \in D$, set $d = q/g$. Every $x \in U_d$ lifts to a unit modulo q, and $13 \mid d$. Equations (5.1) and (5.2) consequently determine the entire layer:

$$f(gx) = \sigma_g F_j, \qquad \sigma_g = \begin{cases} 1, & g = 1, \\ -1, & g > 1, \end{cases} \qquad j = \mathrm{ind}_2(x \bmod 13) \bmod 4. \tag{5.3}$$

The counts used below are recorded in Table 3. Each j-fibre in a listed layer has $\varphi(d)/4$ elements. These layers contain 36552 nonzero residues, leaving only 1472 values of f unrestricted.

TABLE 3. Forced layers for $q = 38025$. The last column is $c_d = 2(2 - v_3(d)) - [3 \mid d]$, the coefficient in twice the prime-3 equation.

g	$d = q/g$	$\varphi(d)/4$	σ_g	c_d
1	38025	4680	1	-1
3	12675	1560	-1	1
5	7605	936	-1	-1
9	4225	780	-1	4
13	2925	360	-1	-1
15	2535	312	-1	1
25	1521	234	-1	-1
39	975	120	-1	1
45	845	156	-1	4

The character equations. For every $d \mid q$ with $13 \mid d$, define the actual fibre sums

$$\Phi_d(j) = \sum_{\substack{x \in U_d \\ \mathrm{ind}_2(x \bmod 13) \equiv j \pmod 4}} f((q/d)x), \qquad j \in \mathbb{Z}/4\mathbb{Z},$$

and put

$$A_j = \sum_{\substack{d \mid q \\ 13 \mid d,\ 5 \nmid d}} \Phi_d(j), \qquad B_j = \sum_{\substack{d \mid q \\ 65 \mid d}} \Phi_d(j).$$

These definitions impose no invariance assumption on f. For each of the three nonprincipal characters of $U_{13}/\langle 3 \rangle \cong C_4$, the character equation in Okada's criterion asserts that the corresponding Fourier coefficient of

$$H_j = 8A_j + 10B_j - 2B_{j+1} \tag{5.4}$$

vanishes. Thus H_j is constant. Indeed, multiply the layered character equation by $(\varphi(13)/13)(3 - 1)(5 - 1) = 96/13$. Since $\chi(3) = 1$, the contribution of the prime 3 is the factor 2 whether or not $3 \mid d$; the contribution of 5 is 4 if $5 \nmid d$ and $5 - \chi(5)$ otherwise. Finally $\mathrm{ind}_2(5) \equiv 1 \pmod 4$, so multiplication of a Fourier coefficient by $\chi(5)$ is translation $j \mapsto j + 1$. This gives (5.4).

By (5.3), the forced part of A_j is $-234F_j$, and the forced part of B_j is

$$(4680 - 1560 - 936 - 780 - 360 - 312 - 120 - 156)F_j = 456F_j.$$

The total cardinality of a j-fibre across the A-layers is

$$\frac{(\varphi(13)+\varphi(169))(1+\varphi(3)+\varphi(9))}{4} = \frac{168\cdot 9}{4} = 378,$$

so at most 144 entries are unrestricted. Across the B-layers the corresponding cardinality is

$$\frac{168\cdot 9(\varphi(5)+\varphi(25))}{4} = 9072.$$

The forced B-fibres contain $4680+4224=8904$ entries, leaving 168 unrestricted entries. We may therefore write

$$A_j = -234F_j + a_j, \quad |a_j|\le 144; \qquad B_j = 456F_j + b_j, \quad |b_j|\le 168.$$

Equation (5.4) becomes

$$H_j = 2688F_j - 912F_{j+1} + e_j, \qquad |e_j| \le 8\cdot 144 + 12\cdot 168 = 3168. \tag{5.5}$$

If the cyclic sign pattern F is nonconstant, it has both a transition $+1\to -1$ and a transition $-1\to +1$. At these two indices the forced expression in (5.5) is respectively 3600 and -3600. Constancy of H would require their difference 7200 to be canceled by an error difference of absolute value at most 6336. This is impossible. Consequently F is constant.

The prime equation. Write $S_d = \sum_{x\in U_d} f((q/d)x)$. Twice Okada's prime-3 equation is

$$\sum_{\substack{d\mid q\\ d>1}} c_d S_d = 0, \qquad c_d = 2(2-v_3(d)) - [3\mid d] \in \{4,1,-1\}.$$

If $S=\sum_{j=0}^{3} F_j$, the contribution of the forced layers is

$$\begin{aligned}(-4680-1560+936-3120+360-312+234-120-624)S\\ = -8886S.\end{aligned}$$

The remaining 1472 signs have coefficients of absolute value at most 4, so their total contribution has absolute value at most 5888. But a constant F has $S=\pm 4$, giving $|8886S| = 35544 > 5888$. This final contradiction proves the theorem. □

Remark 5.3. The argument uses real size bounds in addition to the 2-primary character projection. It does not require parity of fibre sums, an optimization solver, or enumeration of sign patterns. In particular, the existence of bounded odd shadow solutions at a square modulus does not prevent the weighted unit equations from forcing additional layer values that make the resulting real bounds inconsistent.

6. Why bounded orbit sums do not decide the square case

The global theorem excludes all nonsquares. On every remaining odd square, even the exact integer and size conditions on individual orbit sums are feasible. This explains why the additional forcing argument at 195^2 is essential.

Theorem 6.1 (Saturation of the bounded shadow)**.** *For every odd square $q>1$, the 2-shadow equations admit orbit sums satisfying* (2.5)*. In fact every orbit sum can be chosen to have absolute value one. Consequently, some sign function has zero retained shadow, whether or not q is a counterexample modulus.*

Proof. Use the direct square construction in the sufficiency proof of Theorem 3.1. Every coordinate $\Phi_d(y)$ there is 1 or -1, and all retained equations hold. The fibre size

$$h_d = \prod_{p^a\| d} p^{a-1}\frac{p-1}{2^{v_2(p-1)}}$$

is a positive odd integer, so either choice $\Phi_d(y) = \pm 1$ satisfies (2.5). On a fibre with sum Φ and size h, choosing $(h+\Phi)/2$ positive signs and $(h-\Phi)/2$ negative signs produces a sign lift. No assertion about the omitted character equations of that lift is needed. □

Remark 6.2. For fixed feasible orbit sums, the exact number of sign lifts is

$$\prod_{d\mid q,\ d>1}\ \prod_{y\in Q_d} \binom{h_d}{(h_d+\Phi_d(y))/2}.$$

These lifts need not have zero harmonic sum. For example, at $q = 9$ the vector indexed by $0, 1, \dots, 8$,

$$(0, 1, 1, 1, -1, 1, -1, -1, -1),$$

has zero retained shadow. The primitive character modulo 9 with $\chi(2) = e^{2\pi i/6}$, however, gives $\sum_{a\in U_9} f(a)\overline{\chi(a)} = 4$, violating the full conductor-9 equation. Its order is 6, a direction discarded by the 2-shadow. In particular, integer feasibility of the bounded shadow must never be reported as a sign counterexample.

7. A height obstruction to odd-order invariance

The existence of an odd vector in the Okada kernel does not imply the existence of a sign vector in its invariant part. The following theorem gives an infinite family on which this distinction is unavoidable. Here the height of an integer-valued function is its maximum absolute value.

Theorem 7.1 (Large-prime symmetry and height obstruction)**.** *Let $N \geq 1$ be odd, let $r \nmid N$ be an odd prime, let $e \geq 1$, and put $q = Nr^e$ and $m = 2^{v_2(r-1)}$. Suppose that every prime divisor of N is an m-th power residue modulo r. Under the Chinese remainder identification*

$$U_q = U_N \times U_{r^e},$$

let $K = \{1\} \times H_{r^e}$, where H_{r^e} is the subgroup of odd order of U_{r^e}.

Suppose that f is an integer-valued, q-periodic function such that $f(0) = 0$, every $f(n)$ for $n \not\equiv 0 \pmod q$ is odd, f is invariant under multiplication by K, and $L(1, f)$ converges and vanishes. Then

$$N \|f\|_\infty \geq r.$$

Consequently, if $r > N$, there is no K-invariant Erdős function of modulus Nr^e with vanishing sum. In particular there is no such function invariant under the full odd-order subgroup of U_q.

Proof. Set $B = \|f\|_\infty$ and suppose, for a contradiction, that $NB < r$. Let $Q = U_r/H_r$; this cyclic group has order m. Reduction modulo r identifies the quotient U_{r^j}/H_{r^j} with Q, for each $1 \leq j \leq e$.

Every residue $n \pmod q$ with $v_r(n) = e - j$ has a unique representation

$$n = r^{e-j}u, \qquad u \pmod{Nr^j}, \quad r \nmid u.$$

The K-orbits of these residues are indexed by $(a, y) \in (\mathbb{Z}/N\mathbb{Z}) \times Q$, where $a = u \pmod N$ and y is the image of $u \pmod{r^j}$ in Q. Thus K-invariance defines odd integers $b_j(a, y)$ by

$$b_j(a, y) = f(r^{e-j}u), \qquad \Gamma_j(y) = \sum_{a \pmod N} b_j(a, y).$$

Every fibre (a, y) contains

$$h_j = \frac{r^{j-1}(r-1)}{m}$$

residues. Since N is odd, every $\Gamma_j(y)$ is odd; also $|\Gamma_j(y)| \leq NB < r$.

Let $\chi \neq 1$ be a character of Q, viewed as a primitive Dirichlet character of conductor r. The residue hypothesis gives $\chi(p) = 1$ for every prime $p \mid N$. We compute its layered Okada coefficient. Write a divisor containing r as $d = d_0 r^j$, with $d_0 \mid N$. Its weight is

$$\frac{d}{\varphi(d)} \prod_{p \mid d_0} \left(1 - \frac{\chi(p)}{p}\right) = \frac{r}{r-1}.$$

If $n = (q/d)x = r^{e-j}(N/d_0)x$, then $u = (N/d_0)x$ and $\chi(u) = \chi(x)$, again by the residue hypothesis. Summing the layers over all $d_0 \mid N$ therefore sums precisely the residues in the j-th band above. The character equation becomes

$$\begin{aligned} 0 &= \frac{r}{r-1} \sum_{j=1}^{e} h_j \sum_{y \in Q} \Gamma_j(y) \overline{\chi(y)} \\ &= \frac{1}{m} \sum_{y \in Q} \left(\sum_{j=1}^{e} r^j \Gamma_j(y) \right) \overline{\chi(y)}. \end{aligned}$$

The necessity of these equations is the layered Okada criterion. Fourier orthogonality on Q shows that

$$P(y) = \sum_{j=1}^{e} r^{j-1} \Gamma_j(y)$$

is independent of y.

For $y, z \in Q$, the integers $a_j = (\Gamma_j(y) - \Gamma_j(z))/2$ satisfy $|a_j| \leq NB < r$ and $\sum_{j=1}^{e} r^{j-1} a_j = 0$. Reduction modulo r gives $a_1 = 0$; division by r and repetition gives every $a_j = 0$. Thus

$$\Gamma_j(y) = c_j \quad \text{for all } y \in Q,$$

where each c_j is odd and $|c_j| \leq NB < r$.

The remaining nonzero residues are $n = r^e a$, $1 \leq a < N$. Put

$$S_0 = \sum_{a=1}^{N-1} f(r^e a), \qquad |S_0| \leq B(N-1) < r - 1.$$

The strict final inequality follows because NB and r are integers, so $NB \leq r - 1$ and $B \geq 1$. Convergence of a periodic Dirichlet series at 1 implies mean zero, which here reads

$$0 = S_0 + \sum_{j=1}^{e} m h_j c_j = S_0 + (r-1) \sum_{j=1}^{e} r^{j-1} c_j.$$

Consequently $r - 1$ divides S_0, so $S_0 = 0$ and $\sum_{j=1}^{e} r^{j-1} c_j = 0$. Reduction modulo r gives $c_1 = 0$, because $|c_1| < r$, contradicting its oddness. □

Remark 7.2. The proof needs only mean zero and the $m-1$ conductor-r character equations. It does not use any prime-logarithm equation, including the prime-r equation. No converse is claimed: these few equations are necessary, not a sufficient criterion for membership in the full Okada kernel. The theorem concerns odd integer vectors with a height restriction; it does not prohibit odd 2-integral rational vectors of arbitrary height or denominator.

Remark 7.3 (Sharpness when $N = 1$). The height bound is attained at $q = r^2$. Let χ be the quadratic character modulo r and define

$$f(n) = -\chi(n) \quad (r \nmid n), \qquad f(ra) = r\chi(a) \quad (r \nmid a), \qquad f(0) = 0.$$

This is an odd integer vector of height r, invariant under H_{r^2}. Both layer means are zero. The only nonzero character coefficients occur at χ, where their weighted sum is $r \cdot r + r^2 \cdot (-1) = 0$. Hence the layered Okada criterion gives $L(1, f) = 0$ and equality in the bound $\|f\|_\infty \geq r$.

Corollary 7.4 (Infinitely many squares with no invariant sign solution)**.** *Let $r > 11025$ be prime and $r \equiv -1 \pmod{420}$. Set*

$$q = 11025r^2 = (105r)^2.$$

There is no Erdős function of modulus q with vanishing sum that is invariant under the odd-order subgroup of the local r-component. Nevertheless q has an odd Okada vector, by the all-odd-squares theorem, and

$$\frac{\varphi(q)}{q} = \frac{16}{35}\left(1 - \frac{1}{r}\right) < \frac{1}{2}.$$

There are infinitely many such primes r.

Proof. The congruence gives $r \equiv 3 \pmod 4$, so $m = 2$. Also $r \equiv 11 \pmod{12}$ gives $(3/r) = 1$, while quadratic reciprocity gives

$$(5/r) = (r/5) = (-1/5) = 1, \qquad (7/r) = -(r/7) = -(-1/7) = 1.$$

The theorem applies with $N = 11025$ and $e = 2$. Since $\gcd(419, 420) = 1$, Dirichlet's theorem supplies infinitely many primes in this progression. □

The first prime $r > 11025$ in this progression is 13859, giving $q = 2117592488025$. The result does not rule out a non-invariant Erdős counterexample at these moduli. Rather, any counterexample there would have to have a nonzero Fourier component on which the odd-order local subgroup acts nontrivially. It therefore marks a provable limitation of restricting the sign search to the 2-shadow.

8. A complete prime-adic congruence classification

Let ℓ be a prime and put $R_\ell = \mathbb{Z}_{(\ell)}$. Congruences in R_ℓ refer to its residue field.

Theorem 8.1. *For an integer $q > 1$, the following conditions are equivalent:*

(1) *There is a rational-valued function f of period q such that*

$$f(0) = 0, \quad f(n) \in 1 + \ell R_\ell \ (q \nmid n), \quad \sum_{a=1}^{q} f(a) = 0, \quad L(1, f) = 0.$$

(2) *$q \equiv 1 \pmod \ell$ and $\ell \mid v_p(q)$ for every prime $p \mid q$.*

(3) *$q = m^\ell$ for an integer $m \equiv 1 \pmod \ell$.*

For $\ell = 2$ this is precisely the odd-square classification. For larger ℓ it concerns a rational congruence class of coefficients, not the binary sign alphabet. The equivalence of the last two conditions uses $m^\ell \equiv m \pmod \ell$.

Necessity. The mean equation reduced modulo ℓ gives $q - 1 \equiv 0 \pmod \ell$; in particular $\ell \nmid q$. Write $q = \prod_p p^{e_p}$ and partition its prime divisors into

$$P_+ = \{p : \ell \mid p - 1\}, \qquad P_0 = \{p : \ell \nmid p - 1\}.$$

Set $N_p = \ell^{v_\ell(p-1)}$ and let Q_p have order N_p; thus Q_p is trivial for $p \in P_0$. These are the maximal ℓ-quotients of the local unit groups, since $p \neq \ell$. For $D \mid q$, the fibre size is $m_D = \varphi(D)/\prod_{p|D} N_p$.

Average under the subgroup of units of order prime to ℓ and form the fibre sums Φ_D. They satisfy the Okada equations, and $\Phi_D(y) \equiv m_D \pmod \ell$. Their totals are $S_D = \sum_y \Phi_D(y) \equiv \varphi(D) \pmod \ell$. As before, the mean equation converts the prime equation to

$$B_p^* := \sum_{D : p|D} ((p-1)v_p(D) + 1) S_D = 0. \tag{8.1}$$

The elementary divisor-sum identity

$$\sum_{D : p|D} ((p-1)v_p(D) + 1)\varphi(D) = e_p(p-1)q$$

shows immediately that $\ell \mid e_p$ for every $p \in P_0$.

It remains to treat P_+. Write G_S for the sum of the functions Φ_D with support S, on $Q_S = \prod_{p\in S} Q_p$. With $h_p = (p-1)/N_p$ for $p \in P_+$, its constant residue is

$$G_S \equiv \prod_{p\in S\cap P_+} e_p h_p \prod_{b\in S\cap P_0} (b^{e_b}-1) \pmod{\ell}. \tag{8.2}$$

Indeed, the sum of the positive-exponent fibre sizes at a prime in P_+ is $e_p h_p$ modulo ℓ, whereas at $b \in P_0$ it is $\sum_{j=1}^{e_b} \varphi(b^j) = b^{e_b} - 1$. For $p \in P_+$, every S_D with $p \mid D$ belongs to ℓR_ℓ. Equation (1) therefore implies

$$T_p(G) := \sum_{S\ni p} \sum_y G_S(y) \in \ell N_p R_\ell. \tag{8.3}$$

Fold the trivial factors into the nontrivial ones. For nonempty $S \subseteq P_+$ define

$$\widetilde{G}_S = \sum_{U\subseteq P_0} \left(\prod_{b\in U} \frac{b-\tau_b}{b-1} \right) G_{S\cup U}, \qquad (\tau_b F)(y) = F(y+\bar{b}). \tag{8.4}$$

All operators are R_ℓ-integral, commute, preserve total sums, and act as the identity on constant residues. Consequently the normalized character equations regroup exactly into the same system on P_+, and $T_p(\widetilde{G}) = T_p(G)$ for $p \in P_+$. Writing $r = \prod_{b\in P_0} b^{e_b}$, (2) yields

$$\widetilde{G}_S \equiv r \prod_{p\in S} e_p h_p \pmod{\ell}. \tag{8.5}$$

Apply the support-elimination argument of Lemmas 3.2 and 3.3, replacing parity by constant residue modulo ℓ. Here are the necessary checks for this replacement. An anchored difference annihilates a constant residue, so the decomposition into functions missing a coordinate is R_ℓ-integral and each summand has constant residue. The transfer operator is still

$$\frac{N_p}{p-1}(p-\tau_p).$$

Its coefficient is a unit of R_ℓ, and its value on a constant-residue function is zero modulo ℓ, because $p \equiv 1 \pmod{\ell}$. The remaining coordinate set is nonempty and has cardinality divisible by ℓ, so the change in T_p lies in $\ell N_p R_\ell$. Character cancellation is unchanged. This descent uses the character equations, (3), and the constant residues; a separate mean equation for the folded functions is not needed.

After eliminating all supports of size at least two, each singleton is a constant a_p, with $a_p \equiv r e_p h_p \pmod{\ell}$ by (5). Equation (3) gives $N_p a_p \in \ell N_p R_\ell$, hence $a_p \equiv 0 \pmod{\ell}$. Both r and h_p are units, so $\ell \mid e_p$. If P_+ is empty, this step is unnecessary. Necessity follows for every prime divisor. □

Sufficiency. Assume condition 2. First let $p \in P_+$ and $\ell \mid e$. Choose a period-p function u with $u(0) = 0$, $u(1) = 2-p$, and $u(a) = 1$ at all other nonzero residues. It has zero mean and all nonzero-residue values are 1 modulo ℓ. Set

$$c_t = 1 \ (0 \le t < e-1), \qquad c_{e-1} = -\sum_{t=0}^{e-2} p^{e-1-t}.$$

Every $c_t \equiv 1 \pmod{\ell}$ and $\sum_t c_t p^{-t} = 0$. The period-p^e function with value $c_t u(n/p^t)$ when $v_p(n) = t < e$, and zero when $p^e \mid n$, has zero mean and

$$L(1,g) = \left(\sum_{t=0}^{e-1} \frac{c_t}{p^t} \right) L(1,u) = 0.$$

Now put $r = \prod_{b \in P_0} b^{e_b}$. If $r = 1$, omit this factor. Otherwise $r \equiv 1 \pmod{\ell}$, since every prime in P_+ is 1 modulo ℓ. Every $\varphi(D)$ for $D \mid r$ is a unit of R_ℓ, and all its quotient groups are trivial. We construct a function constant on each gcd layer. Start with layer totals $\Phi_D = \varphi(D)$ for $D > 1$. Their mean is $r - 1$ and their prime equations are $B_b^* = e_b(b-1)r$, all divisible by ℓ.

Choose a prime $b_0 \mid r$ and subtract $r - 1$ from Φ_{b_0}. The mean is now zero and the new prime totals are

$$\beta_b = e_b(b-1)r - \mathbf{1}_{b=b_0} b(r-1) \in \ell R_\ell.$$

For each prime $b \mid r$, set $w_b = -\beta_b/(b-1) \in \ell R_\ell$; subtract w_b from Φ_b and add it to Φ_{b^2}. The layers b^2 exist because $e_b \geq \ell \geq 2$. This preserves the mean, changes B_b^* by $(b-1)w_b = -\beta_b$, and leaves every other prime equation unchanged. All resulting Φ_D remain congruent to $\varphi(D)$ modulo ℓ.

Assign the constant value $\Phi_D/\varphi(D)$ to the layer with denominator D. All nonprincipal character coefficients vanish, because the function is constant on each unit layer; the mean and prime equations have just been made zero. Thus the full vanishing criterion gives $L(1, g) = 0$, while every nonzero-residue value is 1 modulo ℓ.

Finally compose the constructed factors: for periods A, B, use $F(n) = u(n)$ if $A \nmid n$ and $F(n) = v(n/A)$ otherwise. The period is AB, the mean remains zero, the congruence class is preserved, and $L(1, F) = L(1, u) + A^{-1}L(1, v)$. This constructs the required function for q. □

Two explicit ternary-congruence examples. For $\ell = 3$, $q = 7^3$, the first construction uses $u(1) = -5$, $u(2) = \cdots = u(6) = 1$ and $(c_0, c_1, c_2) = (1, 1, -56)$. Its Dirichlet-series multiplier is $1+1/7-56/49 = 0$. The second construction applies to $q = 64 = 4^3$, although its prime divisor 2 is not 1 modulo 3. Set

$$f(0) = 0, \qquad f(32) = 196, \qquad f(16) = f(48) = -128,$$

and set $f(a) = 1$ at every other nonzero residue modulo 64. All nonzero-residue values are 1 modulo 3 and the period sum is zero. Every unit equation has value $1+1/2+1/4+1/8-128/16+196/32 = 0$; its prime-2 equation is also zero. Neither example is a binary sign counterexample.

9. Verification, consequences, and remaining questions

The nonsquare obstruction and the exclusion at 195^2 are analytic proofs; neither depends on optimization output or a finite testing range. Their algebra was checked separately, including the character translations and the preservation of the congruences used in the elimination. The sign constructions have a different status: they are explicit finite certificates verified by exact rational arithmetic. The supplement includes compact divisor-layer tables, full value arrays, and standard-library programs that check the original unit and prime equations independently of the discovery model. No small floating-point L-value is used to certify equality to zero.

The original seeds required for the construction are included as data, with exact full-equation checks. A second implementation checks the sparse masks using functional graphs of multiplication on all residues. For 195^2, independent programs check the elementary counts and also rebuild the original equation space before checking its separating certificates. For the global theorem, exact coefficient-level tests of the elimination circuits supplement the proof; they are not a substitute for its induction. These are internal independent implementations and mathematical reviews, not external peer review.

The earlier finite computations on 255 exponent cores supported on $\{3, 5, 7, 11\}$ are retained in the supplement as regression data. They are no longer needed to infer a theorem about nonsquares. Likewise, the constructive exponent-reduction argument is supplied as an additional note: each positive exponent may be reduced to 1 if odd and 2 if even without changing odd-vector existence. The global classification makes that equivalence immediate, but the explicit transformations may be useful for manipulating certificates.

Three questions from the preceding manuscript now have more precise answers. Its general odd-vector conjecture is proved; the question at 195^2 has a negative answer; and sparse masks

construct infinite families beyond product closure. The suggestion that every odd square satisfying $2\varphi(q)+1 \le q$ might work is false, by $q = 195^2$. Determining all counterexample square moduli remains open in this work. In particular, the large-prime symmetry obstruction does not exclude non-invariant sign functions at the moduli it describes. A search which restricts to invariant signs cannot settle those cases.

The same fixed sign alphabet at higher integer arguments does not supply a further open nonvanishing problem: for $k \ge 2$,

$$|L(k,f)| \ge 1 - \sum_{\substack{n\ge 2\\ q\nmid n}} n^{-k} = 2 - (1-q^{-k})\zeta(k) > 0.$$

This is the elementary first-term estimate, not a new result about the Chowla–Milnor conjecture. No unrelated named open conjecture is claimed to be resolved here.

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